Yining Zhu

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Education

Imperial College London London, UK
Master of Science in Mathematics (MSci), First Class Honours Sep 2022 – Sep 2023
Relevant Coursework: Deep Learning, Statistical Learning, Credit Risk Modelling, Stochastic Differential Equations, Stochastic Simulation

Imperial College London London, UK
Bachelor of Science in Mathematics, First Class Honours Sep 2019 – Sep 2022
Relevant Coursework: Machine Learning, Time Series Analysis, Option Pricing, Optimization, Scientific Computation, Applied Probability, PDEs

Experience

Mako Trading London, UK
Graduate Trader Sep 2025 – Present
Brazil Equity Options Trading

- Managed live options market-making books across Brazilian financial names, sizing and timing volatility trades based on IV-RV spreads, skew regimes, dispersion relationships and event calendars while actively managing intraday delta, gamma and vega exposures.
- Built a counterparty volatility-deflection model by regressing signed vega flow (5-minute intervals, top 40 brokers) against ATM volatility changes across 5-minute, 30-minute and 1-hour horizons; identified informed cohorts with 2–3× baseline impact and improved delta edge capture by 50–70% versus production.
- Built a broker trade reconstruction pipeline classifying 10+ multi-leg option structures and aggressor direction, enabling counterparty flow tracking and volatility signal generation.

STIR Prop Trading

- Built a SONIA options probability-density framework combining discrete BoE/ECB meeting distributions with continuous post-meeting rate dynamics, generating probability-weighted policy-path scenarios to identify option strike dislocations and volatility-surface relative-value opportunities; executed delta-neutral strip trades exploiting implied rate-path mispricings.
- Developed intraday relative-value strategies across EURIBOR and SONIA futures using tick-level VWAP data, including cointegration-based calendar-spread mean reversion with walk-forward validation and CPI-driven lead–lag signals between serial and quarterly contracts.

BNP Paribas London, UK
FX Options Trader Intern Jun 2025 – Sep 2025

- Developed a pricing model for USDHKD options using a discrete jump-diffusion framework with up/down/stay scenarios inside the 7.75–7.85 currency peg.
- Built a P&L attribution tool reconciling daily PV changes with theta and other risk factors, improving transparency of desk-level performance analysis.
- Analysed intraday volatility contribution through time-sliced volatility decomposition, providing insights into the distribution of market risk throughout the trading day.

Barclays London, UK
Quantitative Analytics Aug 2023 – May 2025

- Built macroeconomic scenario models using GDP and CPI data to analyse economic growth risk and explain cross-asset return dynamics.
- Applied PCA, LASSO and cross-validation techniques to large-scale financial time-series datasets for factor construction and model selection.
- Performed Monte Carlo simulations to estimate loss distributions and Value-at-Risk metrics.
- Constructed a global growth-factor mimicking portfolio across 14 major asset classes, explaining approximately 40% of GDP forecast deviations.

Research Projects

Proximal Gradient Descent on Convex Optimization London, UK
Master's Thesis Jun 2022 – Aug 2023

- Studied Proximal Gradient Descent and Iterative Shrinkage-Thresholding Algorithms (ISTA) for sparse optimisation and LASSO regression.
- Applied LASSO regularisation within Finite Element Methods to obtain approximate solutions to partial differential equations.

Achievements

- Gold Level, WorldQuant Brain Challenge
- GCSE Mathematics – 1st in the World
- A-Level: A*A*A*

Skills

Programming: Python, SQL, R, Linux, Git, LaTeX
Quantitative: Options Pricing, Time Series Analysis, Machine Learning, Statistical Modelling, Monte Carlo Simulation
Interests: Climbing, Racing, Skating